

Human Development in Nigeria: A Comparative Analysis Against Peer Economies

World Development Indicators (WDI) | 1960 to 2024

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1. Objective / Problem Statement

This project applies a complete data analytics workflow, covering data acquisition, cleaning, modeling, analysis, and visualization, to examine Nigeria's human development standing relative to four peer African economies: Ghana, Kenya, South Africa, and Egypt. The analysis focuses on three development themes: **education**, **health**, and **poverty and labor**, using indicators from the World Bank's World Development Indicators (WDI) database.

The goal is to answer a practical question relevant to policymakers, researchers, and analysts: ***how does Nigeria's progress in human development compare to similar African economies, and where are the most significant gaps?*** Findings are communicated through an interactive Power BI dashboard and this accompanying report.

2. Dataset Description

Source: World Bank, World Development Indicators (WDI), datatopics.worldbank.org/world-development-indicators. The WDI is the World Bank's flagship collection of cross-country development statistics, covering over 200 countries and territories across economic, social, and environmental themes, with data from 1960 to the present.

Files used: *WDIData.csv* (the primary indicator by country by year dataset) and *WDICountry.csv* (country metadata, including region and income group).

Countries Selected

Nigeria (primary focus), compared against Ghana, Kenya, South Africa, and Egypt, Arab Rep., chosen as regionally and economically relevant peer countries within Africa.

Indicators Selected

Theme	Indicator	WDI Code
Education	Adult literacy rate (% ages 15+)	SE.ADT.LITR.ZS
Education	Primary completion rate (%)	SE.PRM.CMPT.ZS
Education	Government expenditure on education (% of GDP)	SE.XPD.TOTL.GD.ZS
Health	Life expectancy at birth (years)	SP.DYN.LE00.IN
Health	Infant mortality rate (per 1,000 live births)	SP.DYN.IMRT.IN
Health	Current health expenditure per capita (current US\$)	SH.XPD.CHEX.PC.CD
Poverty / Labor	Poverty headcount ratio at \$3.00/day (2021 PPP, %)	SI.POV.DDAY
Poverty / Labor	Gini index	SI.POV.GINI
Poverty / Labor	Unemployment, total (% of labor force, ILO modeled)	SL.UEM.TOTL.ZS

Note on Poverty Measure

The World Bank rebased its Purchasing Power Parity (PPP) methodology in 2025, moving the extreme poverty line from \$2.15/day (2017 PPP) to \$3.00/day (2021 PPP). This report uses the updated \$3.00/day (2021 PPP) figures throughout, consistent with the dataset as downloaded.

3. Data Cleaning Process

Step 1: Filter the Raw Data

The raw *WDIData.csv* file (over 200 countries and around 1,500 indicators, in wide format with one column per year from 1960 to 2024) was filtered in Excel down to the 5 selected countries and 9 selected indicators, reducing the working file to 45 country and indicator rows.

Step 2: Unpivot Years

The filtered data was brought into Power BI's Power Query Editor and **unpivoted** from wide format (one column per year) into long format, with a single Year column and a single Value column, to support time series analysis and dynamic filtering.

Step 3: Correct Data Types

Column data types were corrected: Year set to Whole Number, Value set to Decimal Number.

Step 4: Handle Missing Values

For the selected indicators, zero values identified as placeholders for unavailable observations (for example, no country genuinely reports 0% literacy or 0 years of life expectancy) were converted to null using Replace Values, then removed using Remove Empty Rows. This reduced the dataset from roughly 2,700 possible rows to 1,298 rows of genuine reported data.

Step 5: Create a Country Dimension

A separate country dimension table (*WDICountry.csv*) was imported, trimmed to Country Code, Short Name, Region, and Income Group, and filtered to the same 5 countries.

Step 6: Build a Star Schema

A many to one relationship was established between the main indicator table (Fact_Indicators) and the country dimension table (Dim_Country) on Country Code, forming a simple star schema.

4. Analysis Methodology

Analysis was performed using DAX (Data Analysis Expressions) measures within Power BI, rather than simple visual level aggregation, for two reasons. First, the long format structure stacks all 9 indicators into a single Value column, so a naive SUM or AVERAGE would blend indicators with incompatible units and scales. Second, each indicator has different year to year data availability, so a simple approach of taking the latest year present in the table would not reliably return each indicator's true most recent value.

Fixed KPI Measures

One measure per headline indicator (for example Life Expectancy, Literacy Rate), each isolating that specific indicator and returning its value for the latest year with reported data.

Dynamic Comparison Measure

A single measure using *SELECTEDVALUE()* to read whichever indicator is currently chosen in an interactive slicer, then returning that indicator's latest year value per country. This powers one reusable trend chart and one reusable comparison bar chart across all 9 indicators, rather than requiring 9 separate charts.

The dashboard was built across three pages: an **Overview** page focused on Nigeria's own trajectory, a **Comparison** page showing all five countries side by side, and an **Insights and Recommendations** page translating the findings into written conclusions.

5. Key Findings

Indicator (Nigeria)	Value	Latest Year
Life expectancy at birth	54.64 years	2024
Adult literacy rate	70.41%	Latest available
Current health expenditure per capita	US\$66.82	Latest available
Government expenditure on education (% of GDP)	0.32%	2023
Poverty headcount ratio (\$3.00/day, 2021 PPP)	41.80%	2022
Unemployment rate (ILO modeled estimate)	3.06%	Latest available

- Nigeria's life expectancy has risen from approximately 38 years in 1960 to 54.64 years as of 2024, a gain of over 16 years, but it remains the lowest of the five countries compared. Egypt leads at roughly 70 years over the same period.
- South Africa and Kenya both show a pronounced dip in life expectancy through the 1990s and 2000s, consistent with the peak mortality impact of the HIV/AIDS epidemic across Southern and East Africa. Nigeria's trend shows no comparable dip. All five countries show a slight downturn around 2020 and 2021, consistent with COVID-19 mortality impact.
- Nigeria's government expenditure on education has declined sharply, from roughly 3.0% to 3.2% of GDP in the mid 1970s (3.21% in 1974) to just 0.55% by 2012 and 0.32% by 2023, a sustained, multi decade decline of close to 90%.
- On extreme poverty, Kenya has the highest headcount ratio among the five countries at 45.50% (2022), with Nigeria close behind at 41.80%. Both sit far above South Africa (17.40%) and Egypt, which reports a low single digit rate.

6. Insights

Education Investment Collapse

What the data shows: Nigeria's government education expenditure has declined from roughly 3% of GDP in the mid 1970s to under 0.4% by 2023, a sustained, multi decade decline that leaves it far below the UNESCO recommended benchmark of 4% to 6% of GDP, and behind all four peer countries in this analysis.

Why it matters: This chronic underinvestment plausibly links to Nigeria's comparatively lower literacy and primary completion outcomes relative to its peers.

Poverty Gap

What the data shows: Kenya has the highest poverty headcount ratio in this comparison at 45.5% (2022, \$3.00/day 2021 PPP), with Nigeria close behind at 41.8%. Both sit far above South Africa (17.4%), and further still above Egypt's low single digit rate.

Why it matters: This suggests extreme poverty remains a shared, significant challenge in Nigeria and Kenya specifically, not one unique to Nigeria alone, and that GDP growth by itself has not been sufficient to close this gap in either country.

Life Expectancy Trajectory

What the data shows: Nigeria's life expectancy has risen steadily from roughly 38 years in 1960 to 54.6 years as of the latest available year, while Egypt has reached approximately 70 years over the same period.

Why it matters: This demonstrates that past public health investment has produced real, measurable gains, but the pace of improvement, while positive, has not been fast enough to close the historical gap with peer countries.

7. Recommendations

Increase Education Financing

Nigeria's education expenditure (around 0.32% of GDP) is roughly ten times below the UNESCO benchmark. Even a partial move toward that target could meaningfully improve literacy and completion rates.

Recommended action: Gradually increase public education investment toward the UNESCO recommended benchmark of 4% to 6% of GDP.

Target Extreme Poverty Directly

With Nigeria and Kenya both showing poverty rates above 40%, Nigeria could examine targeted poverty interventions used in peer countries, such as cash transfer or subsidy programs, to identify approaches suitable for its own context.

Recommended action: Expand and improve the targeting of poverty focused interventions.

Sustain Health Investment

Life expectancy gains (from 38 to 54.6 years) show that past health investment is working. Continuing this trajectory, especially toward closing the gap with Egypt's approximately 70 years, should remain an active priority rather than a completed goal.

Recommended action: Maintain and expand current levels of health expenditure, with particular attention to the areas driving Egypt's stronger outcomes.

Monitor Cross-Country Lessons

Since no single country in this comparison leads on every indicator (Egypt leads on life expectancy and poverty reduction, but Nigeria and its peers all show gaps elsewhere), policy learning between these five countries could offer mutually useful lessons.

Recommended action: Establish structured channels for comparing policy approaches and outcomes across these five countries on a regular basis.

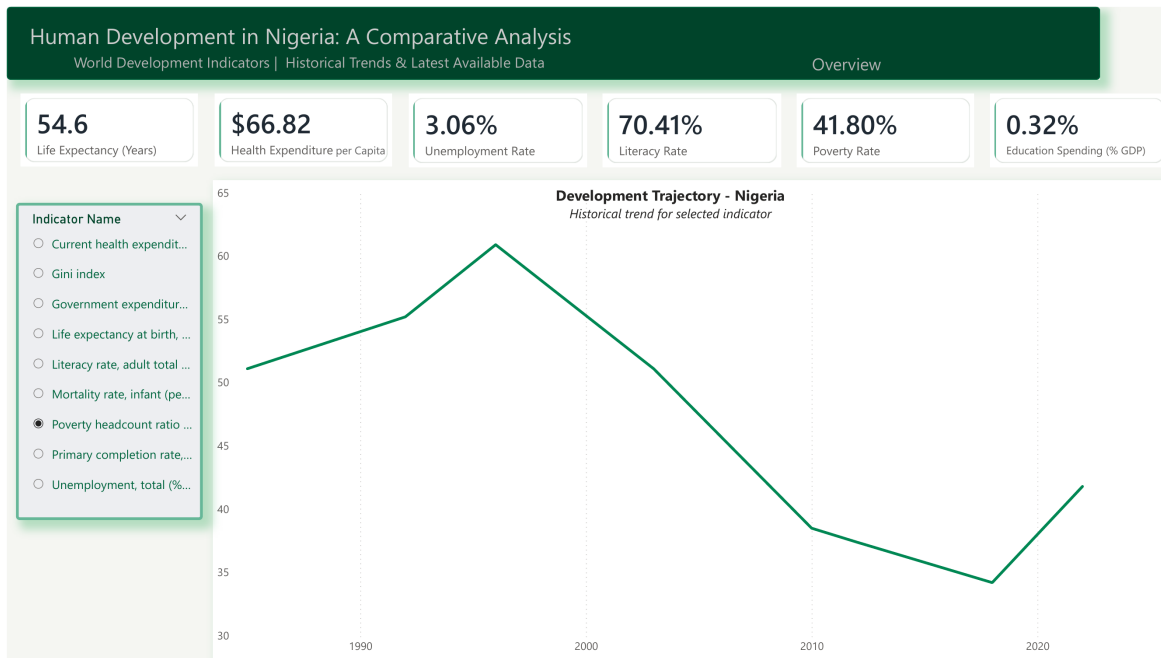
8. Conclusion

This analysis shows that Nigeria has made genuine, measurable progress in human development over the past six decades, most clearly in life expectancy. However, it continues to trail its regional peers on nearly every indicator examined, and shows a uniquely severe, multi decade decline in education investment that stands out even within this comparison group. Extreme poverty remains a shared challenge with Kenya rather than a uniquely Nigerian problem, suggesting that region wide policy approaches, not only country specific ones, may be relevant going forward. The findings highlight the need for sustained, evidence based investment and policy learning if Nigeria is to close its remaining human development gaps.

Supporting materials: interactive Power BI dashboard (.pbix), dashboard screenshots, and project files are available in the accompanying GitHub repository.

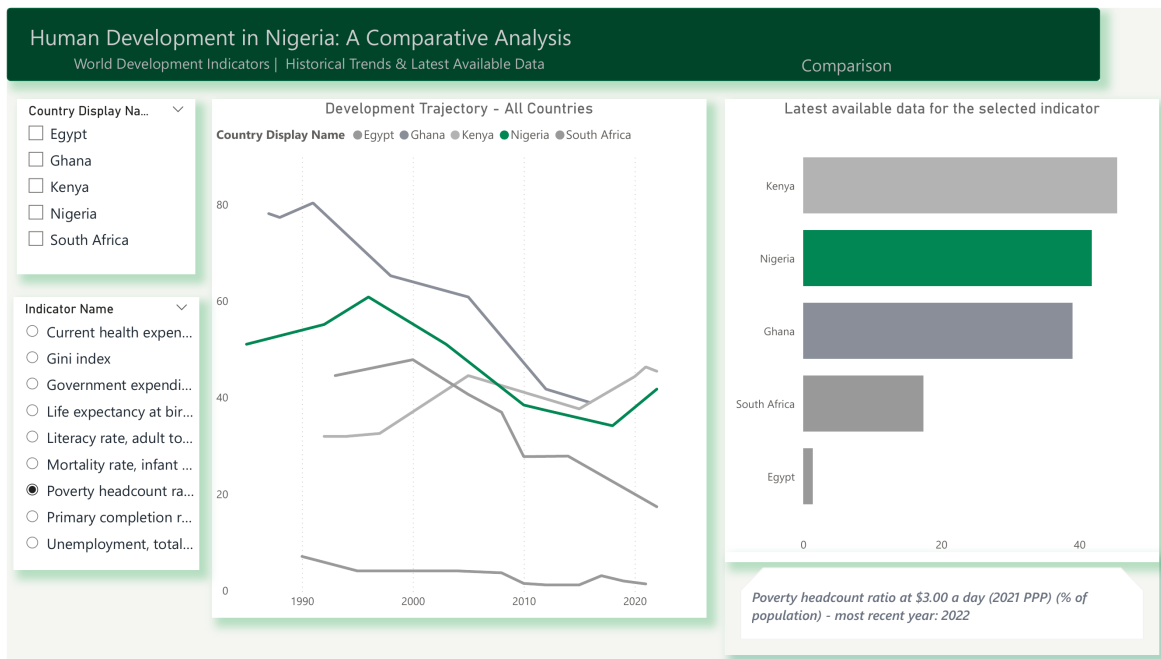
Appendix: Dashboard Screenshots

Overview Page



Nigeria's key indicators and development trajectory, shown with the poverty headcount ratio as the active indicator.

Comparison Page



All five countries compared, with the poverty headcount ratio shown as the active indicator (most recent year: 2022).

Insights and Recommendations Page

Human Development in Nigeria: A Comparative Analysis

World Development Indicators | Historical Trends & Latest Available Data

Insights & Recommendations

Education Investment Collapse

Finding: Education spending declined from approximately 3% of GDP to under 0.4% by 2023

Impact: Nigeria remains below the UNESCO benchmark and behind the peer countries in this analysis.

Poverty gap

Finding: Kenya (45.5%) and Nigeria (41.8%) have the highest poverty rates in the comparison.

Impact: Extreme poverty remains a major shared challenge requiring targeted interventions.

Life Expectancy Trajectory

Finding: Nigeria's life expectancy increased from ~38 years to 54.6 years.

Impact: Progress is evident, but Nigeria still trails all four peer countries with Egypt leading at ~70 years life expectancy.

Increase Education Financing

Gradually increase education spending toward the UNESCO benchmark to support improvements in literacy and completion rates.

Target Extreme Poverty Directly

Strengthen targeted interventions for vulnerable households, particularly in areas with persistent extreme poverty.

Sustain Health Investment

Continue investment in health and development outcomes to sustain life expectancy gains and narrow the gap with peer countries.

Monitor Cross-country Lessons

Study successful approaches across peer countries and adopt relevant policies to Nigeria's context.

Written insights and recommendations derived from the analysis.